H2Former: An Efficient Hierarchical Hybrid Transformer for Medical Image Segmentation

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***Abstract*— Accurate medical image segmentation is of great significance for computer aided diagnosis. Although methods based on convolutional neural networks (CNNs) have achieved good results, it is weak to model the long-range dependencies, which is very important for segmentation task to build global context dependencies. The Transformers can establish long-range dependencies among pixels by self-attention, providing a supplement to the local convolution. In addition, multi-scale feature fusion and feature selection are crucial for medical image segmentation tasks, which is ignored by Transformers. However, it is challenging to directly apply self-attention to CNNs due to the quadratic computational complexity for high-resolution feature maps. Therefore, to integrate the merits of CNNs, multi-scale channel attention and Transformers, we propose an efficient hierarchical hybrid vision Transformer (H2Former) for medical image segmentation. With these merits, the model can be data-efficient for limited medical data regime. The experimental results show that our approach exceeds previous Transformer, CNNs and hybrid methods on three 2D and two 3D medical image segmentation tasks. Moreover, it keeps computational efficiency in model parameters, FLOPs and inference time. For example, H2Former outperforms TransUNet by 2.29% in IoU score on KVASIR-SEG dataset with 30.77% parameters and 59.23% FLOPs.**

***Index Terms*— Hierarchical hybrid transformer, medical image segmentation, multi-scale channel attention, self-attention.**

I. Introduction

PRECISE lesion segmentation from medical images has been one of the fundamental yet challenging problems in medical image analysis, which can benefit disease diagnosis and treatment planning. However, relying on experts for diagnosis is time-consuming and is easily affected by clinical experience. Therefore, automated medical image segmentation is a good auxiliary tool for medical experts.

Recently, convolutional neural networks (CNNs) have been the main-stream in computer vision [1], [2], [3], as well as in the field of medical image analysis [4], [5], [6], [7], [8]. Most of the segmentation methods are based on UNet [7] or its variants [8], [9], [10]. However, due to the intrinsic locality of convolution, CNNs failed to model long-range dependencies, leading to sub-optimal results, and it has the following two shortcomings. Firstly, the small convolutional kernels attend to a local region and force the network to focus on local feature patterns rather than the global features. For medical image segmentation, it often requires long-range global information for reliable segmentation, since the shape and size of lesions vary greatly. Secondly, after training, the convolutional kernels are static and fixed, and the well trained parameters cannot adapt to the input image contents. Therefore, CNNs lack flexibility for different inputs with different characteristics.

Meanwhile, Transformer [11] based on self-attention has achieved remarkable sucsess in natural language processing. It is well known that visual attention plays an important role in computer vision [12], [13]. Therefore, a natural idea is to introduce the self-attention into computer vision. Vision Transformers (ViT) [14] is the first attempt to adopt Transformer encoder for image classification, achieving competitive performance with its CNN counterparts. Because of its great success, Transformers are also popular in other vision tasks, such as object detection [15] and semantic segmentation [16], and achieves the state-of-the-art performance. With the long-range information interaction and dynamic feature encoding ability, Transformers have been applied for medical image segmentation. TransUNet [17] is the first attempt to employ Transformer for medical image segmentation, which is a hybrid structure of CNNs and Transformers. Then, a pure Transformer encoder-decoder structure named Swin-UNet [18] and nnFormer [19] for volumetric medical image segmentation are proposed, showing strong performance.

Although long-range dependencies can be modeled well in Transformer, it still has drawbacks and limits the performance. Firstly, the spatial information is ignored, as it serializes images into 1D tokens, and it is weak in local feature learning, which is crucial for 2D images. Although this problem can be relieved by position encoding, while position encoding needs to adapt to the various input resolution through interpolation, thus affecting the performance. The second drawback of

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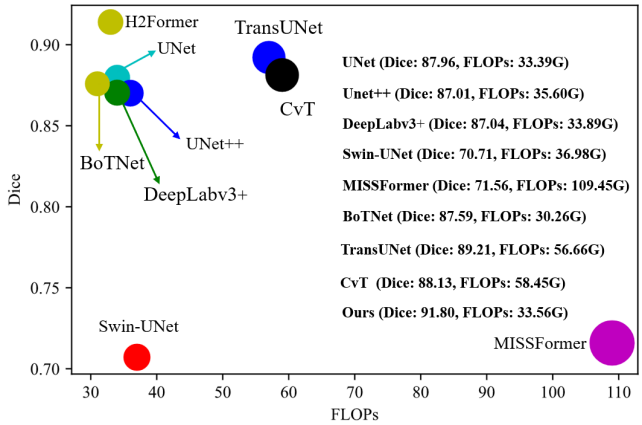

Fig. 1. The Dice score on KVASIR-SEG test dataset compared to other methods with respect to FLOPs (based on ResNet-34 [1]), and it shows that ours method achieves the best trade-off between performance and model complexity. The size of markers represents the FLOPs.

Transformer is the dependence on large-scale datasets with quadratic computational complexity, which can be interpreted as low inductive bias [20] in modeling local visual cues [21]. Finally, Transformer only learns features in a token-wise attention manner with single scale and cannot perceive multi-scale channel-wise feature dependencies, which is harmful for lesions with various shapes and scales. Therefore, there is still room for improvement in hybrid structure of CNNs and Transformers.

From the above analysis, CNNs can preserve the spatial information of features by convolutional kernels with strong inductive bias, which is beneficial to 2D images. It can reduce the requirement for large scale datasets and thus is data-efficient and suitable for medical image segmentation. While Transformer is capable of performing dynamic attention and modeling long-range dependencies with sufficient receptive field, which is helpful in segmenting large scale lesions. Considering the complementary roles of CNNs and Transformer, there are some works [22], [23], [24] to integrate the two feature encoding strategies, while they maintain quadratic computational complexity for high-resolution feature maps, and they are dependent on large scale dataset like ImageNet. Therefore, it is still a problem whether the hybrid structure trained on limited medical images can outperform the performance of pure CNNs and pure Transformer models. Beyond that, most existing hybrid methods only rely on single token granularity within each attention layer in a token-wise manner, and they are incapable of capturing features at different scales. Furthermore, they ignore the channel relationship of the Transformer, which is also important for feature learning with channel attention. Multi-scale features has been demonstrated to be effective in various vision tasks owing to their powerful feature representation ability [25], [26], [27]. The hybrid of CNNs and Transformers with multi-scale features within a layer has not been fully explored in previous network design for medical image segmentation with limited labeled data. Naturally, we seek to ask the question: how to beneficially integrate the merits of CNNs, multi-scale channel attention and Transformers for medical image segmentation while avoiding their respective limitations and keep computation efficiency?

To answer this question, we redesign the network structure, and propose an efficient hybrid Transformer block by incorporating the strong inductive bias of CNNs and multi-scale channel-wise attention with Transformer's token-wise attention, which can aggregate feature extraction capabilities from different aspects. Finally, it forms a hierarchical hybrid vision Transformer (H2Former) for medical image segmentation, which is an encoder-decoder structure. From a macro point of view, the outputs of each stage constitute the multi-scale features, and features within each stage also constitute multi-scale features by our MSCA from a micro point of view, forming the nested multi-scale feature representation. The experimental results demonstrate that H2Former can deliver highly competitive performance with reasonable parameters and floating point operations (FLOPs), as shown in Fig. 1. The main contribution of our method is that we consider multi-scale, local features of CNN and long-range dependency of Transformer within a block, forming a hierarchical hybrid structure with multi-scale information in the stage-level and network-level, and it is important for segmentation task with different shapes and scales. From a macro point of view, the output of each stage forms the multi-scale features. From a micro point of view, each stage also forms multi-scale features by our MSCA, forming the nested multi-scale feature representation. Moreover, we also consider multi-scale feature selection by channel attention, which is ignored by previous works. Because there is a large gap between medical images and natural images, and the labeled training data is limited, it is necessary to elegantly integrate these existing key findings into a unified framework, and we have conducted extensive experiments on both 2D and 3D medical image segmentation tasks to validate the effectiveness of our method. In contrast to the existing approaches, our method provides the following three main contributions.

- We proposed a hierarchical hybrid model that elegantly integrate the local information of the CNNs, multi-scale channel attention features and long-range features of Transformer within a unified block, which can integrate the merits of them simultaneously, and enhanced the feature representation ability of the model.
- A light-weight multi-scale channel attention (MSCA) branch is presented, which serializes the feature maps into multi-scale token pyramids, and then they are calibrated with channel-wise attention. MSCA benefits medical image segmentation task with different shapes and scales and is complementary to token-wise self-attention.
- Finally, we demonstrate the superiority of our method comprehensively in terms of performance, model parameters, FLOPs and inference time, which outperforms the competing models on three 2D and two 3D medical image segmentation tasks, which demonstrates the effectiveness of our model.

II. Related Work

A. CNNs for Medical Image Segmentation

Due to the powerful feature representation capability, medical image segmentation is mainly based on CNNs in the last

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few years. UNet [7] is seminal and widely used for medical image segmentation and has achieved state-of-the-art results on various medical image segmentation tasks [4], [5], [6]. Then, many variants of UNet are developed, such as Dense-UNet [28], Attention UNet [9], UNet++ [8], nnUNet [10] and UNet3+ [29]. Inspired by these fundamental architectures, some methods are tailored for specific tasks, such as optic disc and optic cup segmentation from fundus images [5] and COVID-19 lung infection segmentation [30], etc. All in all, CNNs have made great progress in medical image segmentation.

B. Vision Transformers for Medical Image Segmentation

Recently, the Transformer based methods have shown promising performance for vision tasks [16], [24], [25], [31], [32], [33], and they have been viewed as an alternative to the CNN counterparts. ViT [11] is the first work to apply pure Transformers for image classification, which takes the input image as a sequence of image patches and then adopt Transformer blocks to encode image features. In order to train ViT more effectively, DeiT [34] proposed several efficient training strategies that allow ViT to converge well with no external data. To mitigate the quadratic computational overhead, a window-based Transformer named Swin-transformer [31] was proposed to performs self-attention through local windows for computer vision tasks, reducing computation cost and achieving good performance. In order to introduce desirable properties of CNNs to the Transformers, some works learn from CNN's design philosophy [23], [35], and achieved a strong performance with efficiency.

Due to the promising performance of Transformers, some works have been proposed for both 2D and 3D medical image segmentation [36], [37], [38], and nnFormer [19] is a strong 3D volumetric medical image segmentation method with local and global volume-based self-attention mechanism to learn volume representations. TransUNet [17] is the first to employ a hybrid CNN-Transformer architecture for medical image segmentation. It leverage the local features from CNNs and the global context features from Transformers in a cascade manner, which is a strong alternative to CNNs. Inspired by Swin-transformer [31], Swin-UNet [18] was proposed with a pure Transformer structure. However, it ignore local spatial information, which is extremely essential for segmentation tasks. In order to addresses the dilemma that Transformer requires large amounts of data, a hybrid Transformer architecture (UTNet) [39] that integrates self-attention into a CNN for enhancing medical image segmentation.

C. Analysis of Previous Work

Although the above methods have achieved good results, they still ignored the integration of multi-scale channel attention and hierarchical hybrid manner. Therefore, the problem is how to aggregate the strengths of CNNs, multi-scale channel attention and Transformers to build a unified block with reasonable parameters and computation complexity for medical image segmentation. **Firstly**, for CNNs, they can extract the local spatial information well, but they still have an inevitably limited receptive field and lack the ability of modeling long-range cues. **Secondly**, for Transformers, although long-range dependencies can be modeled well, they still ignore the local detailed information and structure priors to some extent, which is essential for segmentation tasks. And beyond that, pure Transformer models ignore channel-wise attention, and the global self-attention like ViT [11] require large-scale datasets and high computational complexity, making it difficult to train transformers efficiently for medical segmentation applications. **Thirdly**, for the hybrid Transformer, they only consider token-wise self-attention to build the relationship among tokens. It does not fully consider the channel-wise interaction and conduct feature learning with single scale. There are three key differences between the proposed H2Former and previous works, as shown in Fig. 2: (1) Multi-scale feature fusion is crucial to feature representation, and we want to fuse multi-scale feature fusion in Transformer in a simple and efficient way. Although there are previous works like CrossFormer considering multi-scale feature fusion within a stage, they only concatenate these multi-scale features without feature selection, resulting in sub-optimal solutions (See Table VI E.6 and E.7) because of the conflict between features with different scales. Therefore, how to perform multi-scale feature fusion and considering feature selection becomes very important. For this reason, we propose MSCA to conduct dynamic selection of multi-scale features and model the relationship between channel dimensions, which complement token-wise self-attention. (2) The labeled training data is limited in medical domain, and it means that we should fully explore the features from the backbone. Our hybrid block can model the local and long-range dependency at the same time, and multiscale feature with feature selection is also considered to perform more effective multi-scale feature fusion. These merits make the proposed method superior to other hybrid transformer-CNN models using multi-scale feature fusion. Other hybrid transformer-CNN models using multi-scale feature fusion only consider the inherent multi-scale feature from multi-stage and fuse them together without feature selection. In addition to considering this implicit multi-scale feature, we also explicitly modeled the multi-scale feature inside each stage, forming a nested multi-scale feature representation, which is conducive to the network to make full use of the features of various granularity. (3) Previous models only consider the performance of the model or the efficiency of the model, and we further consider both of them at the same time, and thus our model can achieve excellent performance while maintaining a high inference speed (See Table VII and Fig. 1). The experimental results show that such integration can significantly improve the feature representation ability of the model.

III. Methodology

In this section, we first present the overview of our proposed method. Then, the details of the proposed hierarchical hybrid Transformer block will be given.

A. Overview and Preliminary

The overall pipeline of our proposed method is illustrated in Fig. 3, following the design philosophy of [7], [31], the

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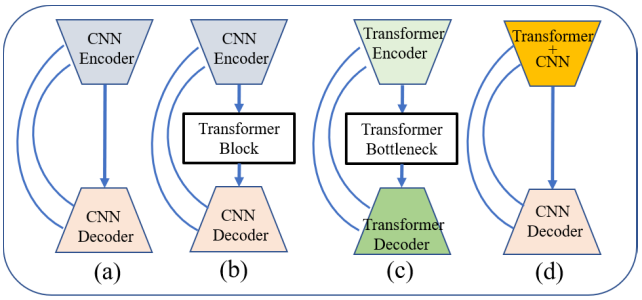

Fig. 2. Conceptual comparison of four current popular methods for medical image segmentation. (a): The standard pure convolutional UNet [7]. (b): The cascaded hybrid of CNN and Transformer structure (*e.g.,* TransUNet [17]). (c): The pure Transformer structure (*e.g.,* Swin-UNet [18] and MISSFormer [40]). (d): Our proposed efficient hierarchical hybrid Transformer structure.

structure consists of four stages in encoder. The main novelty comes from the unified hierarchical hybrid of convolution, multi-sacle channel attention and Transformer feature encoding strategies with computational efficiency. Here, the term "hierarchical" means that we adopt the hybrid Transformer blocks in each stage, which facilitates the medical image segmentation task. The "hybrid" refers to the three feature encoding strategies can be integrated into a unified block.

Specifically, given an input image $I \in \mathbb{R}^{H\times W\times C}$, H, W and C denote the height, width and the channel number of the image, respectively. First, following ViT [11] and Swin-transformer [31], a convolutional stem is used for patch partition, which is implemented with a convolutional layer and a MaxPooling layer. The image is divided into N patches $X = \{x_i \in \mathbb{R}^{P\times P\times C}, i = 1, 2, \ldots, N\}$ of patch size $P \times P$ ($P = 4$ in our paper), and $N = \frac{H\times W}{P^2}$, x_i is the i-th patch. Then, we embed the patches $X \in \mathbb{R}^{N\times(P\times P\times C)}$ to image tokens $E \in \mathbb{R}^{N\times C_1}$ with a trainable linear projection. Then, the embedded tokens E are fed into the four encoder stages to produce the hierarchical features, and the spatial resolution of feature maps in the four stages are with {1/4, 1/8, 1/16, 1/32} of the original input resolution, and these feature maps are also used for skip connections in the decoder, which will provide the fine-to-coarse features for segmentation. We denote the output features of stage$_i$ as $F_i \in \mathbb{R}^{\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}\times C_i}$, $i \in \{1, 2, 3, 4\}$, note that $F_0 = E$. Patch merging layer is used to reduce the number of tokens and expand the token dimensions between two adjacent stages, which is implemented with a 3×3 convolutional layer of stride 2 for the token number reduction. Each stage consists of one hybrid Transformer block (Sec. III-B), which can capture multi-scale long-range features and local spatial information with feature selection. Then, the hierarchical features F_i of the four stages are fed into the patch expanding path (Sec. III-C) to predict the segmentation results $Y \in \mathbb{R}^{H\times W\times K}$, where K is the number of categories. Next, we elaborate the proposed network design.

B. Hybrid Transformer Block

Due to the intrinsic locality of convolution, which is incapable of modeling long-range dependencies. In addition, Transformer generates single scale features with only token-wise attention, and it ignores the relationship among channels, thus subpar to tackle situations such as segmenting multi scale lesion regions in medical images. Considering these issues, we introduce the hybrid Transformer block during the feature extraction, which consists of the multi-scale channel attention, a convolutional layers and Transformer layers, forming a unified block, as shown in Fig. 3(b).

1) Multi-Scale Channel Attention (MSCA): Multi-scale and channel-wise relationships are very important for medical image segmentation with differen shapes and scales, and thus we propose multi-scale channel attention to enable attention mechanism in a channel-wise perspective to capture multi-scale dependencies of tokens, which is a complement to token-wise self-attention in Transformer.

As we can observe in Fig. 4, The multi-scale tokens have the same tokenization center but with different context scales, and all these tokens will be concatenated to form multi-scale tokens. In practice, the process can be implemented by convolutional layers with different kernels with stride of 1, note that we do not reduce the number of tokens in the feature maps. The multi scale features of stage$_i$ are denoted as $\hat{F}_i^m$, $i \in \{1, 2, 3, 4\}$, which can be obtained by:

$$\hat{F}_i^m = \mathrm{Concat}(T_i^s), \quad s \in S = \{2, 4, 8, 16, 32\}, \tag{1}$$

where $\hat{F}_i^m \in \mathbb{R}^{\frac{HW}{2^{2(i+1)}}\times C_i}$ is the multi scale tokens in stage$_i$, Concat denotes the token concatenation. $T_i^s \in \mathbb{R}^{\frac{HW}{2^{2(i+1)}}\times C_i^s}$ is the tokens with scale of s and dimensions of C_i^s in stage$_i$, satisfying $\sum_{s\in S} C_i^s = C_i$, and T_i^s can be obtained by:

$$T_i^s = \mathrm{MST}_i^s(F_{i-1}), \quad i \in \{1, 2, 3, 4\}, \tag{2}$$

where MST_i^s is the multi-scale tokenization layers with $s \times s$ kernels and stride of 1 in stage$_i$. And then, the multi-scale token features can be calibrated by channel attention:

$$F_i^m = \sigma(\mathrm{Conv1d}(\mathrm{GAP}(\hat{F}_i^m))) \otimes \hat{F}_i^m, \quad i \in \{1, 2, 3, 4\}, \tag{3}$$

GAP is the global average pooling layer to aggregate features from different scales. Conv1d is the fast 1D convolutional layer with kernel size of 3 according to [41] to perform channel relationship. σ is a Sigmoid activation to get the weights that indicate the importance of each channel. Then, multi-scale features are calibrated via the multiplication of channel weights and original features. $F_i^m \in \mathbb{R}^{\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}\times C_i}$ is the final calibrated multi-scale features.

To reduce the computational complexity of the MSCA, lower dimension is used for large scales while high dimension for small scales. Assuming that the dimension of input tokens is C_i in stage$_i$, for the first stage of high resolution feature maps with low dimension, we adopt four scales (*i.e.*, 4×4, 8×8, 16×16 and 32×32), and the output dimension are $C_i/2$, $C_i/4$, $C_i/8$ and $C_i/8$, respectively, making it the same dimensions as the input, as shown in Fig. 4. For the rest of low resolution feature maps, there is no need to adopt large kernel and we use two scales of 2×2 and 4×4 (See Table VI), and the output dimensions are $C_i/2$ and $C_i/2$. Therefore, MSCA can extract multi-scale features with reasonable computational complexity.

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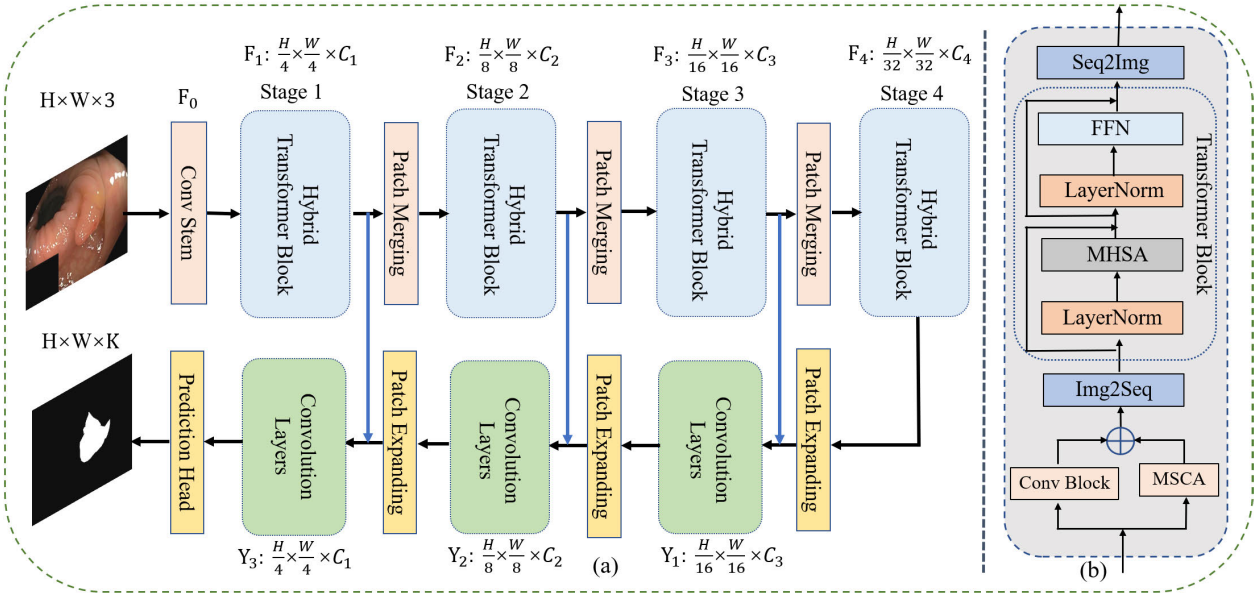

Fig. 3. (a) Overall architecture of the proposed H2Former. The encoder consists of four stages, and each stage contains a hybrid Transformer block. (b) The structure of the proposed hybrid Transformer block. It extracted local features and multi-scale channel attention features through the convolutional block and MSCA, and then aggregates them together. Then, rich features are extracted by the Transformer block to achieve the purpose of combining the advantages of CNNs, multi-scale channel-wise attention and token-wise features of Transformers simultaneously. 'Img2Seq' and 'Seq2Img' denote the reshape operation to meet the shape requirements.

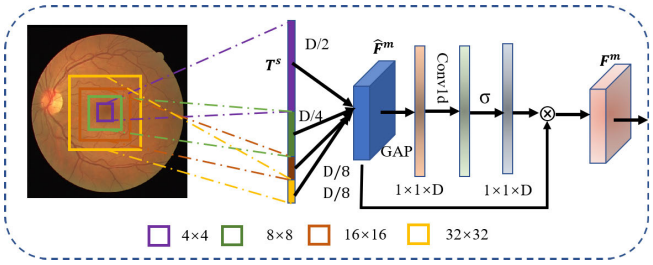

Fig. 4. The schematic diagram of the proposed multi-scale channel attention (MSCA), which consists of four scales with (4 × 4, 8 × 8, 16 × 16 and 32 × 32) and perform channel-wise attention among different channels of tokens.

2) Convolutional Block: To better encode spatial location information and inject strong inductive bias, we adopt convolutional block to extract local spatial features. Specifically, given an input features F_{i-1}, we adopt the convolutional block to model local spatial features, which are shown as follows:

$$F_i^c = \text{ConvBlock}_i(F_{i-1}), \quad i \in \{1, 2, 3, 4\}, \tag{4}$$

$F_i^c \in \mathbb{R}^{\frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times C_i}$ is the local features, which contains 2D spatial location information, making it possible to encode position information. Since we focus on the macro hybrid of CNNs and Transformers rather than the micro network structure design, and thus we adopt the off-the-shelf convolutional network structures. The convolutional blocks in the four stages are the same as $Conv_2$, $Conv3$, $Conv4$ and $Conv5$ in original ResNet-34 [1] architecture.

3) Transformer Block: The structure of Transformer block is shown in the upper part of Fig. 1(b), which consists of a Multi-Head Self-Attention (MHSA) layer to build long-range dependencies, two Layer Normalization and a Feed-Forward Network (FFN). To make Transformer block be aware of multi-scale channel dependencies and local dependencies, the extracted multi-scale channel-wise token features F_i^m is integrated with local position features F_i^c as the input of Transformer block:

$$X_i = \text{Img2Seq}(F_i^m + F_i^c), \tag{5}$$

The output of the Transformer block in the i-th stage can be formulated as:

$$F_i' = \text{MHSA}(\text{LayerNorm}(X_i)) + X_i, \tag{6}$$

$$F_i = \text{Seq2Img}(\text{FFN}(\text{LayerNorm}(F_i')) + F_i'), \tag{7}$$

where Img2Seq is the reshape operation to change the 2D feature maps to 1D tokens, and Seq2Img is the reverse operation. LayerNorm is the Layer Normalization [42] and F_i is the output of the i-th encoder stage. MHSA is the window based multi-head self-attention using regular or shifted window partitioning [31] performed within each window, which is formulated as:

$$\text{MHSA}(x) = \text{Softmax}(Q(x)K(x)^T/\sqrt{d})V(x), \tag{8}$$

Q, K, V are the query, key and value transformation matrices, and they are implemented with three linear projection layers; d is the dimension of input tokens. The self-attention is the token-wise attention, which is complementary to multi-scale channel-wise attention. FFN can be formulated as:

$$\text{FFN}(x) = \text{Linear}(\text{GELU}(\text{Linear}(x))), \tag{9}$$

where x is the features from the window-based self-attention, Linear and GELU is the linear projection layer and non-linear activation function, respectively. Following [17], [31], [40], the expansion ratio of the first linear layer in FFN is set to 4, and

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the window sizes for window self-attention is 16×16. For each stage, the number of transformer block is $\{2, 2, 2, 2\}$, and the number of head in the four stages is set to $\{2, 4, 8, 16\}$.

The hybrid Transformer block can model local spatial relationships and introduce the inductive bias of CNNs, and this property makes the network data-efficient, which is suitable for medical image segmentation with low data regime. In addition, it can also model the multi-scale channel-wise and token-wise long-range dependencies at the same time, which integrates the advantages of them into a unified block. There are two main advantages between our hybrid Transformer block and previous hybrid structure: 1) Our hybrid Transformer block introduces hierarchical spatial information, global feature encoding and the multi-scale channel attention mechanism into a unified block to capture multi-granularity channel-wise and token-wise features, which can better model lesion features with different sizes and shapes in medical images; 2) It achieves better segmentation results in both 2D and 3D medical image segmentation tasks, and maintains reasonable parameters and FLOPs. By combining the three kinds of features, our method can learn better feature representation.

C. Patch Expanding Path

We Observe That Patch Expanding With Pure Transformer Structure Like Swin-UNet [18] and MISSFormer [40] Will Leads to Two Problems: First, it introduces redundant computational cost during training and testing. Through the experimental results in Table VI, we find that the simple convolutional decoder can get pretty good segmentation results, because the encoder has strong ability to extract rich features for decoder. The second problem is that redundant features will cause difficulties in optimization, thus confusing the model and causing sub-optimal solutions. Therefore, we adopt the simple convolution for patch expanding with efficiency.

In the patch expanding path, we adopt convolutional layers for decoding the extracted hybrid multi-scale features $\{F_1, F_2, F_3, F_4\}$ in the encoder path. We upsample the token features by patch expanding (PE) layer, which is implemented as a transposed convolution layer. After patch expanding, we concatenated the features from encoder path with the same resolution like skip connection in UNet:

$$Y_i = \mathrm{Conv}_i(\mathrm{Concat}(\mathrm{PE}_i(Y_{i-1}), F_{4-i})), \quad (i = 1, 2, 3), \quad (10)$$

where Y_i denotes the output features of each decoder stage, $Y_0 = F_4$, Conv denotes the convolutional layers consists of a 3×3 convolutional layer, a Batch Normalization [43] layer and a ReLU activation layer, and Concat denotes the channel-wise concatenation. In the final prediction head, Y_3 are firstly resized to the same resolution as the input, and then we apply a 1×1 convolution layer followed by the softmax activation function to map them to K channels, corresponding to (K-1) lesion maps and one background map.

For training, the supervised loss is defined as:

$$\mathcal{L} = \mathcal{L}_{ce}(y, p) + \lambda \mathcal{L}_{dice}(y, p), \quad (11)$$

where $\mathcal{L}_{ce}$ is the cross-entropy loss and $\mathcal{L}_{dice}$ is the Dice loss, y and p are the ground truth and prediction results, respectively. Due to the imbalanced issue in fundus lesion segmentation, the combination of the two loss function can effectively alleviate the problem of class imbalance, and the default setting of λ is 1.

IV. Experiments

In this section, we evaluate the method on three 2D and two 3D medical image segmentation tasks. We first introduce the benchmark datasets, implementation details and evaluation metrics. Then we compare our method with state-of-the-art methods and perform ablation studies to validate the design choices of the proposed architecture.

A. Datasets

1) IDRiD Dataset [44]: The Indian Diabetic Retinopathy Image Dataset (IDRiD) is available for the segmentation and grading of retinal image that hosted by International Symposium on Biomedical Imaging (ISBI) conference 2018. This fundus lesion segmentation dataset consists of 81 images with the resolution of 2848×4288, and pixel-wise label masks of hard exudates (EX), hemorrhages (HE), microaneurysms (MA) and soft exudates (SE) are provided. Due to the small fundus lesions, there is a serious imbalance between background and foreground lesions, which is a great challenge for pixel-level segmentation tasks. There is an official split of 54 images for training and 27 images for testing.

2) Kvasir-SEG Dataset [45]: This is a dataset for gastrointestinal polyp segmentation, which contains 1000 polyp images and their corresponding label masks. Since there is no official train and test dataset split, we perform 5-fold cross validation and report their average results.

3) Skin Lesion Dataset [46]: This is a dermoscopic image analysis benchmark challenge for automated diagnosis of skin cancer. The dermoscopy image dataset we used is the ISIC 2016 skin lesion segmentation challenge dataset. It contains 900 training dermoscopic images and 379 test images, and the binary ground truth masks are labeled by the experts.

4) ACDC Dataset [47]: Automatic Cardiac Diagnosis Challenge (ACDC) is a dataset for automated cardiac diagnosis, which conains 100 patients. For all these data, the corresponding manual annnotations for the cavity of the right ventricle (RV), the myocardium (MYO) of the left ventricle and the cavity of the left ventricle (LV) are given by clinical experts. The dataset is split into 70 training samples, 10 validation samples and 20 test samples, the same as nnFormer [19].

5) Synapse Dataset [48]: This dataset consists of 30 CT scans for multi-organ segmentation, which conains 13 organs, which are spleen, right kidney, left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal vein and splenic vein, pancreas, right adrenal gland and left adrenal gland. The dataset is split into 20 training samples and 10 test samples, and the evaluation setting is the same as [19].

B. Implementation Details and Evaluation Metrics

1) Implementation Details: In the experiments, to reduce overfitting and improve the generalization ability of the model,

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we adopted data augmentation of horizontal flips, vertical flips, and random rescales. the network was trained for 90 epochs using the AdamW [49] optimizer and the weight decay is 0.0001. The initial learning rates is 0.0001, which is reduced by the "Poly" strategy. The number of feature channels for the four stages are [64, 128, 256, 512]. For polyp and skin lesion segmentation, we resized the images and labels to 512 × 512, and the batch size is 18. To preserve the details of tiny lesions in fundus images of IDRiD dataset, the input size is set to 960 × 960 and the batch size is 3. For ACDC and Synapse datasets, we resized the 2D slices and labels to 320 × 320 and 480 × 480, respectively, and the batch size is 18. The initial learning rates is set to 0.0005 and 0.0007, respectively. The Transformer blocks in our proposed model are randomly initialized, and the convolutional layers are pretrained on ImageNet [50]. Note that all the experiments were conducted with the same experiment settings for a fair comparison and we did not use any pre-processing or post-processing. Our framework was implemented with Pytorch and all experiments were performed on an NVIDIA GeForce RTX 3090 GPU. Code will be available at https://github.com/NKUhealong/H2Former.

2) Evaluation Metrics: The following commonly used metrics are adopted for systematical performance evaluation, including pixel-wise Accuracy (Acc), Mean Absolute Error (MAE), Dice Similariy Coefficient (Dice), and Intersection-over-Union (IoU).

C. Comparisons With Other Methods

To demonstrate the effectiveness of our proposed method, we compare it against its CNN, Transformer and hybrid counterparts on both 2D and 3D medical image segmentation tasks. To make fair comparisons, we obtained the results with convincing parameters for these models. For CNN and hybrid methods, we adopt ResNet-34 pre-trained on the ImageNet as backbone to keep reasonable parameters and computational cost except SFA [51], PraNet [52] and ACSNet [53], and the three results are obtained from their papers.

1) Results on Skin Lesion Segmentation: The quantitative results on skin lesion segmentation task are shown in Table I. Compared to these models, our method outperforms the three kinds of counterparts by a large margin, which achieves the best trade-off between performance and model complexity. To be more specific, our H2Former achieves 92.41% Dice and 86.45% IoU, surpassing TransUnet by 1.10% and 1.49% with only about 33.71M parameters, which is about 31% of TransUnet with about 109.54M parameters. Moreover, our method also surpasses the SOTA nnUNet and nnFormer, as they are designed systematically for 3D volumetric datasets, may not be suitable for 2D segmentation. The superior performance owns to the hybrid Transformer block that makes it possible to process spatial local information and multi-scale long-range dependencies efficiently within a unified block. From the results, we find that the hierarchical hybrid model has distinct advantages over the pure Transformer, pure CNN methods and naive cascaded methods, which is more friendly to medical image segmentation tasks with a small number of training samples.

TABLE I
QUANTITATIVE COMPARISONS TO PREVIOUS STATE-OF-THE-ART METHODS ON THE ISIC 2016 SKIN LESION DATASET. WE CONDUCT EXPERIMENTS FOR FIVE TIMES AND REPORT THE AVERAGE RESULTS

Type	Method	MAE↓	Acc↑	Dice↑	IoU↑	#Param↓
CNN	UNet [7]	4.98	94.52	89.56	83.61	23.63M
	UNet++ [8]	5.12	94.21	89.46	83.63	24.38M
	Att-UNet [9]	4.89	94.75	89.25	83.35	23.72M
	PSPNet [26]	4.28	95.54	90.22	83.87	14.26M
	DeepLabv3+ [27]	4.42	95.43	90.12	84.35	26.19M
	SFA [51]	4.72	94.78	89.74	83.97	-
	PraNet [52]	4.68	94.87	90.20	84.73	-
	ACSNet [53]	4.83	94.99	90.14	84.21	-
	nnUNet [10]	4.35	95.59	90.45	84.52	-
Trans	Swin-UNet [18]	4.74	95.03	90.12	83.21	27.27M
	nnFormer [19]	4.83	95.17	90.08	83.18	-
	MISSFormer [40]	5.45	94.36	88.37	80.75	42.33M
Hybrid	ResT [24]	5.09	94.91	89.41	82.61	34.79M
	BoTNet [23]	4.46	95.52	90.19	83.54	14.87M
	TransUNet [17]	4.23	95.75	91.31	84.96	109.54M
	CvT [22]	4.63	94.56	90.42	84.25	32.23M
	H2Former(Ours)	**3.80**	**96.31**	**92.41**	**86.45**	33.71M

TABLE II
QUANTITATIVE COMPARISONS TO PREVIOUS STATE-OF-THE-ART METHODS ON THE KVASIR-SEG DATASET. WE CONDUCT 5-FOLD CROSS VALIDATION AND REPORT THEIR AVERAGE RESULTS

Type	Method	MAE↓	Acc↑	Dice↑	IoU↑	#Param↓
CNN	UNet [7]	4.21	95.62	87.96	82.34	23.63M
	UNet++ [8]	4.56	95.51	87.01	78.84	24.38M
	Att-UNet [9]	4.42	95.49	87.14	82.06	23.72M
	PSPNet [26]	4.14	95.68	87.95	82.62	14.26M
	DeepLabv3+ [27]	4.16	95.71	87.04	82.05	26.19M
	SFA [51]	7.50	-	72.30	61.10	-
	PraNet [52]	3.00	-	89.80	84.00	-
	ACSNet [53]	3.00	-	85.15	78.67	-
	nnUNet [10]	3.96	96.03	89.75	83.59	-
Trans	Swin-UNet [18]	6.63	93.31	70.71	60.96	27.27M
	nnFormer [19]	4.12	95.98	89.15	83.06	-
	MISSFormer [40]	7.11	92.98	71.56	61.17	42.33M
Hybrid	ResT [24]	6.45	92.79	86.21	79.54	34.79M
	BoTNet [23]	4.39	95.23	87.59	82.61	14.87M
	TransUNet [17]	3.52	96.40	89.21	83.73	109.54M
	CvT [22]	4.03	95.73	88.13	82.04	32.23M
	H2Former(Ours)	**2.52**	**97.49**	**91.80**	**86.29**	33.71M

2) Results on Polyp Segmentation: Then, we conducted experiments on polyp lesion segmentation task. Similarly, as shown in Table II, the proposed method still outperforms other competitive methods. Specifically, our method is better than the best method PraNet [26] by 2.00% in Dice and 2.29% in IoU. Note that TransUNet adopts Transformer in the last stage of CNNs, which ignores the hierachical and multi-scale features and thus obtain sub-optimal performance. The results demonstrate the effectiveness and benefits of our proposed hierarchical hybrid architecture.

3) Results on IDRiD Lesion Segmentation: To further verify the capacity of our method, we conduct the tiny fundus lesion segmentation experiments on IDRiD dataset, which contains four small lesion types. Due to the imbalanced issule in

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TABLE III
QUANTITATIVE COMPARISONS TO PREVIOUS STATE-OF-THE-ART METHODS ON THE IDRiD DATASET. WE CONDUCT EXPERIMENTS FOR FIVE TIMES AND REPORT THE AVERAGE RESULTS

Type	Method	Dice↑	IoU↑	AUC↑	#Param↓
CNN	UNet [7]	52.12	40.13	41.09	23.63M
	UNet++ [8]	53.26	41.32	39.14	24.38M
	Att-UNet [9]	53.01	40.23	41.75	23.72M
	PSPNet [26]	51.52	39.56	41.09	14.26M
	DeepLabv3+ [27]	53.22	41.14	56.53	26.19M
	SFA [51]	52.04	40.79	43.29	-
	PraNet [52]	53.21	41.56	47.54	-
	ACSNet [53]	52.34	41.04	42.78	-
	nnUNet [10]	52.03	40.61	49.86	-
Trans	Swin-UNet [18]	53.21	41.43	56.68	27.27M
	nnFormer [19]	52.13	40.69	54.63	-
	MISSFormer [40]	51.79	39.86	58.37	42.33M
Hybrid	ResT [24]	45.23	34.20	55.67	34.79M
	BoTNet [23]	53.61	41.06	57.74	14.87M
	TransUNet [17]	52.21	39.86	53.74	109.54M
	CvT [22]	51.35	39.79	54.45	32.23M
	H2Former(Ours)	**57.16**	**44.85**	**69.03**	33.71M

TABLE IV
QUANTITATIVE COMPARISONS TO PREVIOUS STATE-OF-THE-ART METHODS ON THE ACDC DATASET. THE EVALUATION METRIC IS DICE SCORE

Type	Method	Average↑	RV↑	MYO↑	LV↑
CNN	UNet [7]	89.41	87.77	85.88	94.67
	UNet++ [8]	89.58	87.23	87.13	94.37
	Att-UNet [9]	89.01	87.30	85.07	94.66
	PSPNet [26]	88.75	85.99	86.39	93.87
	DeepLabv3+ [27]	88.25	85.41	85.44	93.90
	SFA [51]	89.24	86.42	87.83	93.49
	PraNet [52]	90.16	87.21	88.73	94.54
	ACSNet [53]	89.39	86.94	87.04	94.21
	nnUNet [10]	91.61	90.24	89.24	95.36
Trans	nnFormer [19]	92.06	90.94	89.58	95.65
	Swin-UNet [18]	90.00	88.55	85.62	**95.83**
	MISSFormer [40]	84.53	81.07	81.21	91.29
	UNETR [54]	88.61	85.29	86.52	94.02
Hybrid	LeViT-UNet [55]	90.32	89.55	87.64	93.76
	BoTNet [23]	88.75	85.99	86.39	93.87
	TransUNet [17]	89.71	88.86	84.54	95.73
	CvT [22]	89.01	87.30	85.07	94.66
	H2Former(Ours)	**92.40**	**91.31**	**90.12**	95.76

fundus lesion segmentation task, MAE and Acc metrics are not suitable to measure the segmentation performance, and thus we adopt Dice and IoU to measure the segmentation performance. We report the average performance of the four lesion types in Table III, extensive experiments shows that our method can outperform the previous methods by a large margin, and reveal the great potential of our method in tiny lesion segmentation tasks.

Following TransUNet [17], our method are trained with 2D slices and tested in a slice-by-slice fashion for all the 3D volumes, and then the predicted 2D slices are stacked together to reconstruct the 3D prediction results for evaluation, and thus it can compare with recent competing methods fairly. For the evaluation metric, we adopt Dice score and it is the same as nnUNet [10] and nnFormer [19].

4) Results on ACDC for Automated Cardiac Segmentation: As shown in Table IV, we can see that the best methods is H2Former, and it was better than nnUNet and nnFormer on Dice score with naive training strategies. While nnUNet and nnFormer are specifically designed for 3D medical image segmentation, and the training strategy is also specially optimized for nnUNet. While our method adopt naive training recipe without pre-processing, post-processing and advanced strategies like deep supervision, which demonstrates the effectiveness of our approach.

5) Results on Multi-Organ Segmentation: Because this dataset has 13 classes, different from the dataset used in nnFormer [19] and nnUNet [10], the dataset they used only contains 8 classes, and thus we re-implemented thes methods with their released code. We report the average performance of the 13 multi-organ segmentation results, and the 95% Hausdorff Distance (HD95) and Dice score are shown in Table V. Compared with the current SOTA method like nnUNet and nnFormer, our method can achieve the best results, which

TABLE V
QUANTITATIVE COMPARISONS TO PREVIOUS STATE-OF-THE-ART METHODS ON THE MULTI-ORGAN SEGMENTATION DATASET

Type	Method	Dice↑	HD95↓
CNN	UNet [7]	67.89	26.60
	UNet++ [8]	68.50	42.39
	Att-UNet [9]	67.40	35.73
	PSPNet [26]	67.74	30.28
	DeepLabv3+ [27]	66.53	29.58
	SFA [51]	67.43	26.94
	PraNet [52]	68.79	21.49
	ACSNet [53]	68.04	24.52
	nnUNet [10]	69.67	23.58
Trans	nnFormer [19]	69.76	20.55
	Swin-UNet [18]	67.74	20.28
	MISSFormer [40]	66.50	24.36
Hybrid	BoTNet [23]	65.98	36.92
	TransUNet [17]	68.04	27.21
	CvT [22]	67.11	21.77
	H2Former(Ours)	**70.52**	**15.03**

shows that our proposed hybrid Transformer block is still effective for 3D medical image segmentation.

The results of H2Former on the three 2D and two 3D medical image segmentation datasets fully demonstrate its excellent ability of handling different modalities with large and tiny organs or lesions with reasonable model complexity. Comprehensive experimental results show that the proposed hierarchical hybrid vision Transformer can achieve superior performance against other SOTA methods in quantitative results for different medical image segmentation tasks.

6) Visualization Results: For qualitative analysis, we show the 2D visual comparisons of our method and other models in Fig. 5 and Fig. 6. It can be seen that the predictions of H2Former captures the features really well and generate much better segmentation results. For more challenging cases, such

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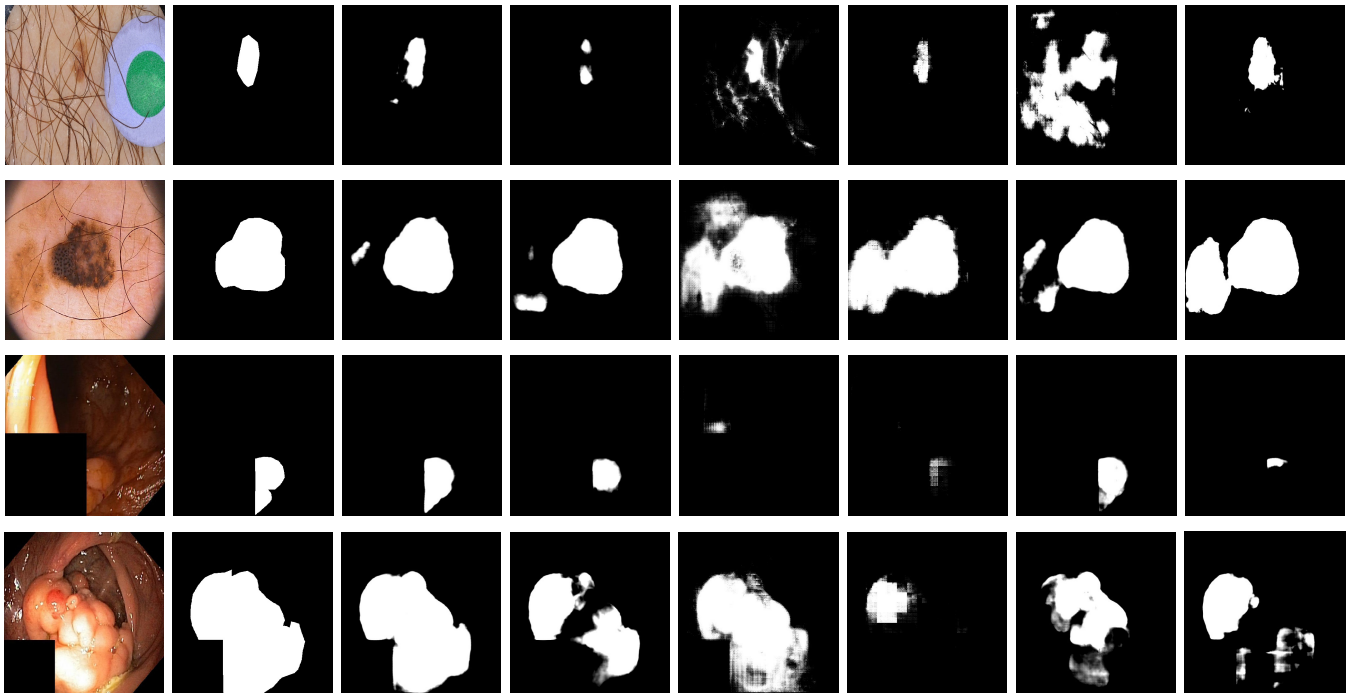
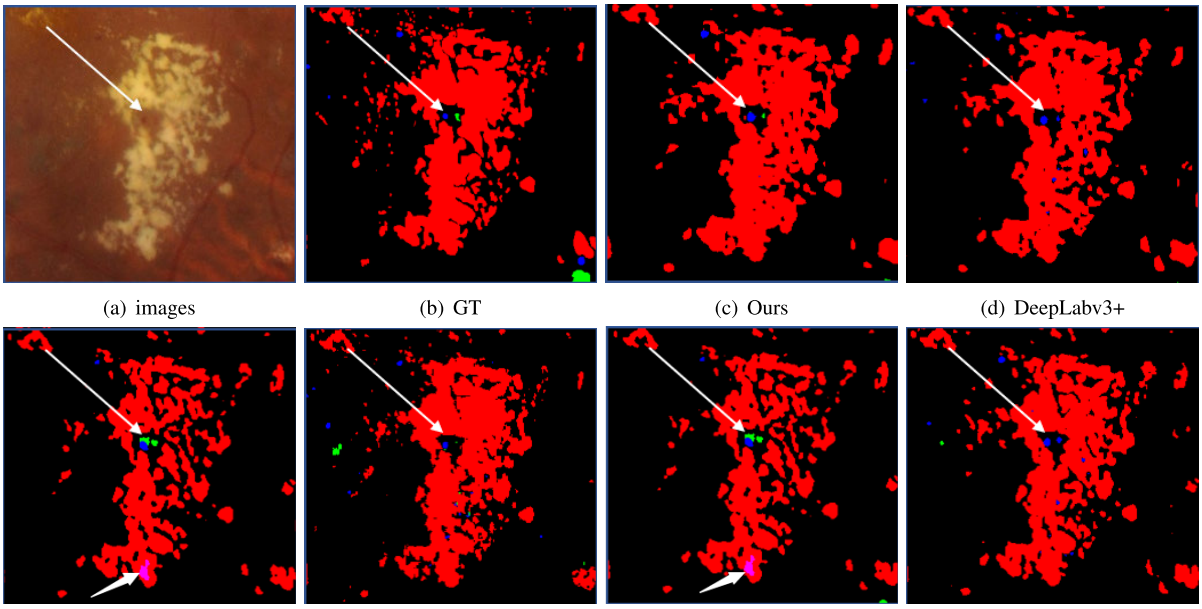


(a) images (b) GT (c) Ours (d) DeepLabv3+ (e) MISSFormer (f) Swin-UNet (g) TransUNet (h) UNet

Fig. 5. The visual comparison results of skin and polyp segmentation tasks. The images in the first two rows are the results of skin lesion segmentation, and the last two rows are the results of polyp segmentation. It can be seen that skin image obscured by the hair and polyp image with the complex surrounding tissues can be well handled.



(a) images (b) GT (c) Ours (d) DeepLabv3+

(e) MISSFormer (f) Swin-UNet (g) TransUNet (h) UNet

Fig. 6. The visual comparison results of fundus lesion segmentation. As indicated by the white arrows, our model is able to segment small lesions in the fundus images under low contrast, while other methods could not segment them well, showing that our method could not only segment large lesions, but also small lesions with strong robustness. Red, green, blue, and pinkmarkings denote hard exudates, hemorrhages, microaneurysms and soft exudates, respectively.

as skin image in the first row obscured by the hair and polyp image in the third row with the complex surrounding mucosa, our model handles these challenging cases well and can produce clear boundaries. The segmentation performance of other models will be decreased due to the challenging occlusion and complex surrounding mucosa. As shown in Fig. 6, for the small lesion segmentation from fundus images, H2Former is still capable of segmenting the tiny lesions indicated by the white arrows. We can observe that the small lesions highlighted by the white arrows are misclassified by other

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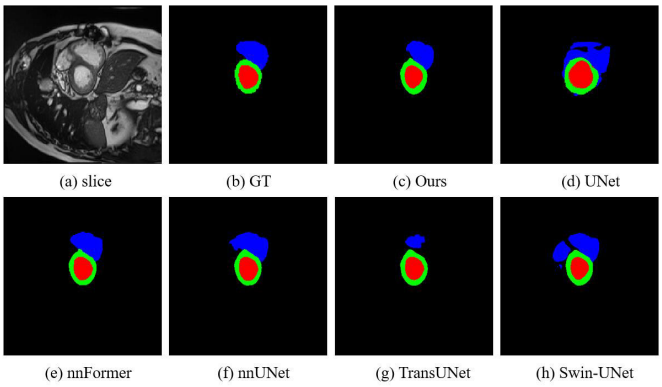

Fig. 7. Visualization results of automatic cardiac segmentation on ACDC dataset, red, green, and blue denote the LV, Myo and RV, respectively.

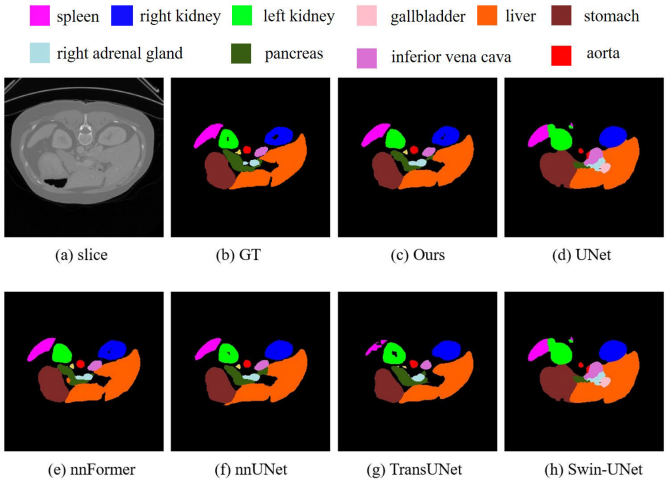

Fig. 8. Visualization results of automatic multi-organ segmentation on Synapse dataset.

models. As the hybrid Transformer blocks can encode local features and long range pixel-wise dependencies, which makes our predictions more precise and better than other models. For cardiac and multi-organ segmentation, the visual results are shown in Fig. 7 and Fig. 8, we can see that our method produces very competitive segmentation results compared with other SOTA methods. Especially, for multi-organ segmentation task, our method is superior to other methods in dealing with the boundary of organs, and has better capability on segmenting organs. These results verify that our model works well with 3D volumetric data in a 2D slice way.

D. Ablation Studies

We conduct a series of ablation studies to investigate the effectiveness of our proposed framework and justify the design choices on Kvasir-SEG dataset, and the baseline method is UNet based on ResNet-34 [1]. Unless otherwise stated, the decoder of all the settings remains the same, and we perform 5-fold cross validation and report their average results in Table VI.

1) Hierachical Hybrid Encoder: By comparing E.4 against E.1, E.2 and E.3, we can see that CNNs combined with Transformer in a hierachical hybrid manner significantly outperforms pure CNN encoder, pure Transformer encoder and cascaded manner, which shows that CNNs and Transformer can indeed be effectively incorporated in terms of encoding local spatial features and long-range dependencies, and thus they are complementary to each other, achieving promising hybrid results, which improve the baseline with 2.44% in IoU. It suggests that local details and long-range dependencies are both crucial to medical image segmentation in the hierachical manner.

TABLE VI
ABLATION STUDIES ON KVASIR-SEG DATASET. 'TRANS' DENOTES TRANSFORMER, 'CASC' DENOTES CASCADED MANNER AND 'HIER' DENOTES HIERACHICAL MANNER

Setting	Method	MAE↓	Acc↑	Dice↑	IoU↑	FLOPs↓
E.1	Pure CNN encoder	3.57	96.28	88.41	82.73	24.39G
E.2	Pure Trans encoder	4.51	95.39	86.46	80.36	24.21G
E.3	CNNs+Trans(casc)	3.45	96.48	89.60	83.71	48.36G
E.4	CNNs+Trans(hier)	3.11	96.80	90.61	85.17	31.96G
E.5	H2Former+Trans Decoder	3.09	97.02	90.85	85.13	42.36G
E.6	CNNs+Trans(hier)+CEL	2.98	97.08	91.04	85.23	33.05G
E.7	CNNs+Trans(hier)+MSCA	**2.52**	**97.49**	**91.80**	**86.29**	33.56G
E.8	H2Former+four scales	2.68	97.38	91.74	86.07	34.73G
E.9	H2Former+linear embed	2.71	97.08	91.35	86.03	33.21G

2) Multi-Scale Channel Attention (MSCA): By comparing E.4 and E.7, the results show that MSCA can bring better performance, improving the Dice score and IoU score by 1.19% and 1.12%, respectively, which indicates that multi-scale channel attention can make up for the shortcomings of single scale token-wise self-attention, and it models the relationship among different channels, which benefits the lesion segmentation with different scales and shapes. In order to validate the importance of multi-scale feature selection, we make a comparison with CEL layer in CrossFormer [25], and set the stride as 1. It can be seen from E.6 and E.7 that the results without feature selection are inferior to our results, which indicates that it is necessary to consider feature selection when performing multi-scale feature fusion. In order to verify that there is no need to adopt large scales for the feature maps in stage 2, stage 3 and stage 4, four scales are used in multi-scale channel attention (2×2, 4×4, 8×8 and 16×16). It can be found from E.7 and E.8 that the large scales are not helpful for low-resolution feature maps, but increases the computation cost. The reason is that for the late stages, after the large multi-scale feature extraction of the first stage, their receptive fields are sufficient, and thus two small scales of 2×2 and 4×4 are enough to achieve a good performance.

3) Hybrid Manner: For the hybrid choice, there are two options: one is to integrate Transformer blocks directly in the final stage of ResNet-34 like TransUNet [17] (cascaded), or the other is to interact them at each feature level in a hierachical manner. By comparing E.4 and E.3, we can observe that cascaded manner yields sub-optimal performance because it ignores multi level information interaction, which drops the IoU by 1.46%, and thus the hierachical manner is better than cascaded. We believe that such short and long-range feature interaction can play a role of feature constraint, that

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is, the two kinds of features will not deviate from each other. However, cascade manner may lead to deviation between shallow short-range features and deep long-range features, resulting in performance degradation.

4) Convolution Stem: To verify the necessity of the convolution stem, reshape and linear embedding are used instead of convolution stem for patch partition, from the ablation study results of E.9 and E.7, it can be found that the performance of convolution stem is better than that of reshape and linear embedding. The reason is that convolution stem can provide strong local spatial information, while linear embedding will not preserve the local position information well, thus convolution stem performs better. Therefore, we use a convolution stem for patch partition.

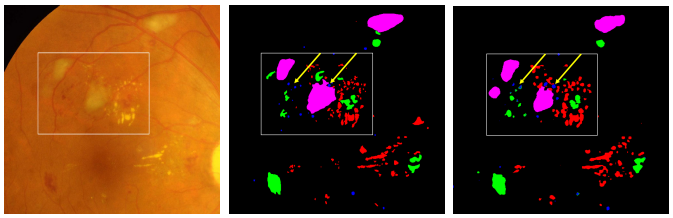
5) Final Structure: Further, by comparing E.7 (H2Former) against other settings, we observe that it achieves the best results among them, and outperforms baseline (E.1) by 3.39% in Dice and 3.56% in IoU, which means that it is necessary to consider the long-range dependencies, local spatial information and multi-scale tokenization features within a unified block in medical image segmentation. As shown in E.5, if we replace the convolutional decoder with Transformer decoder, the performance deteriorates and the computational cost increases significantly. Compared with CNN, Transformer lacks the perception ability of 2D spatial information, and thus it failed to recover the spatial information well when the decoder partially recovers the original spatial position information, while CNN has a strong inductive bias and strong perception ability of position information, and it can better recover the position information and obtain accurate segmentation mask. The Transformer decoder is more complex than the convolutional decoder, it requires more training data to fit. However, the labeled medical images are limited and not enough to train the over-complex network structure, and thus the simple convolutional decoder works better. Therefore, the light-weight convolutional decoder is more efficient than complex Transformer decoder, and it can achieve satisfactory results with reasonable complexity.

E. Model Complexity

In this section, we compare the complexity of the proposed model with other methods, including parameters, FLOPs and inference time. FLOPs and inference time are calculated based on the input of 512 × 512 on an NVIDIA GeForce RTX 3090 GPU using the PyTorch framework. As shown in Table VII, MISSFormer has 42.33M parameters and 109.45G FLOPs, which is 3.26 times of our method, this is because it adopts pure Transformer structure with global self-attention and causes a large amount of computational cost. Our method has 33.71M parameters with 33.56G FLOPs, which is only 30.77% and 59.23% of TransUNet respectively, and the segmentation performance is obviously better than it. Although Swin-UNet has fewer parameters, its encoder and decoder both adopt transformer structure, resulting in increased computational cost and its performance is inferior to our method. For UNet, UNet++, Att-UNet and DeepLabv3+, although their parameters is similar to ours, their FLOPs is comparable or higher than ours. Further, their segmentation performance is significantly worse than our method, see Table I, Table II and Table III. Therefore, our hybrid structure can ensure good results while keep reasonable computational cost, and thus it is a more effective structure. From the results in Table VII, our model achieves the best trade-off between performance and model complexity in terms of the number of parameters, FLOPs and the inference time.

TABLE VII
ANALYSIS OF PARAMETERS, FLOPS AND INFERENCE TIME FOR VARIOUS MODELS WITH INPUT SIZE OF 512 × 512. AND THE BACKBONES ARE BASED ON RESNET-34 [1]

Type	Method	#Params↓	FLOPs↓	Inference Time↓
CNN	UNet [7]	23.63M	33.39G	28.87 ms
	UNet++ [8]	24.38M	35.60G	31.30 ms
	Att-UNet [9]	23.72M	33.93G	29.49 ms
	PSPNet [26]	14.26M	57.88G	50.39 ms
	DeepLabv3+ [27]	26.19M	33.89G	29.62 ms
Trans	Swin-UNet [18]	27.27M	36.98G	34.82 ms
	MISSFormer [40]	42.33M	109.45G	92.86 ms
Hybrid	BoTNet [23]	14.87M	30.26G	23.87 ms
	TransUNet [17]	109.54M	56.66G	48.65 ms
	CvT [22]	32.33M	58.45G	49.71 ms
	H2Former(Ours)	33.71M	33.56G	29.02 ms



(a) images (b) GT (c) Ours

Fig. 9. A failure case from IDRiD dataset. As we can see from the segmentation results, there are confusions among different lesion types and the missed small lesions. It is very challenging to segment lesions in such a complex context because four types of lesions appear in the adjacent region and the low contrast of this region.

F. Failure Cases Analysis

It is also important to analyze the limitations of the model, and thus we show a failure case of fundus lesion segmentation in Fig. 9. We can see that the lesion regions indicated by the white rectangle are highly similar to the surrounding regions with low contrast, the segmentation performance will be decreased. And some lesions in fundus image are very small, only a few tens of pixels, and thus it is easy to be overlooked during downsampling, which makes it difficult to segment. The segmentation results show the misclassification, blurred boundary and the neglect of small lesions, as shown by the yellow arrows. The main reason for this failure case is that features are difficult to be retained and differentiated for this challenging region, resulting in poor segmentation performance for tiny lesions. First, a simple method is to increase the resolution of the input images and improve the perception ability of the model for small lesions, but this will have high requirements for GPU memory and computational cost, which may not be practical. Second, designing more

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powerful self-attention models could improve the segmentation results, more advanced training piplines like self-supervised learning and semi-supervised learning may enhance the feature diversity and improve the generalization ability of the model.

V. Conclusion and Future Work

In this work, to leverage the benefits of CNNs, multi-scale channel-wise and token-wise features of Transformer, we proposed a hierarchical hybrid vision Transformer for medical image segmentation, dubbed as H2Former, which integrates the inductive bias of CNNs, multi-scale channel attention token features and the Transformer to combine the merits of them into a unified block. As an alternative framework to the CNNs, H2Former can increase the model's capability for modeling multi-scale long-range dependencies and local spatial information. It is superior to the previous CNNs, Transformer and hybrid methods, showing better results on several 2D and 3D medical image segmentation tasks without any advanced training strategies. It also demonstrates that the proposed method achieves good trade-off in the model complexity and segmentation performance.

In the future, we plan to validate the capabilities of H2Former on other tasks such as medical image classification, self-supervised learning and semi-supervised learning.

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